

University of Pisa
Department of Mathematics
Master's Degree in Mathematics

Algebraic Number Theory 3

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1. p-adic number fields

Definition 1.0.1 (Brauer group)

Let k be a field and k_s be one of its separable closures. The *Brauer group* of k is

$$\mathrm{Br}(k) := H^2(k, k_s)$$

If $K|k$ is a separable subextension of $k_s|k$ we can define the *relative Brauer group* of the extension as

$$\mathrm{Br}(K|k) := \ker\left(\mathrm{Br}(k) \xrightarrow{\mathrm{Res}} \mathrm{Br}(K)\right)$$

The elements of $\mathrm{Br}(k)$ correspond to the finite dimensional division algebras D with center k. In this correspondence, the subgroup $\mathrm{Br}(K|k)$ gives those D for which $D \otimes_k K \simeq M_{n\times n}(K)$, where $n^2 = \dim_k D$.

Lemma 1.0.1

1. If $K|k$ is a Galois extension with group G then it holds $\mathrm{Br}(K|k) \simeq H^2(G, K^\times)$;
2. The Brauer group $\mathrm{Br}(k)$ is the union of all the subgroups $\mathrm{Br}(K|k)$ for finite Galois extensions $K|k$;
3. If G is finite and cyclic it holds $\mathrm{Br}(K|k) \simeq \frac{k^\times}{N_{K|k}(K^\times)}$.

Here $N_{K|k} : K^\times \to k^\times$ denotes the norm of the field extension. When $K|k$ is finite and Galois it has value

$$N_{K|k}(a) = \prod_{\sigma\in \mathrm{Gal}(K|k)} \sigma a$$

for every $a \in K^\times$.

Proof:

1. From Hilbert's theorem 90 we have $H^1(G, K^\times) = 0$. As seen in # this makes the following inflation-restriction sequence exact:

   $$0 \longrightarrow H^2(G, K^\times) \xrightarrow{\mathrm{Inf}} H^2(k, k_s) \xrightarrow{\mathrm{Res}} H^2(K, k_s),$$

   and by definition $\mathrm{Br}(K|k) = \ker \mathrm{Res} = H^2(G, K^\times)$.
2. From the previous point we deduce

   $$\mathrm{Br}(k) = \varinjlim H^2(G, K^\times) \overset{(1)}{=} \varinjlim \mathrm{Br}(K|k) = \bigcup \mathrm{Br}(K|k).$$
3. As seen in # when the Galois group G is cyclic we have

   $$\mathrm{Br}(K|k) \overset{(1)}{=} H^2(G, K^\times) = \frac{(K^\times)^G}{N_{K|k}(K^\times)} = \frac{k^\times}{N_{K|k}(K^\times)}.$$

□

Example 1.0.1

0. When the field k is complete we have $\mathrm{Br}(k) = 0$ since the Galois action on it is trivial.
1. Let's compute the Brauer group of the real numbers:

$$\mathrm{Br}(\mathbb{R}) = \mathrm{Br}(\mathbb{C}|\mathbb{R}) = H^2(\mathrm{Gal}(\mathbb{C}|\mathbb{R}), \mathbb{C}^\times) = \frac{\mathbb{R}^\times}{N_{\mathbb{C}|\mathbb{R}}(\mathbb{C}^\times)} = \frac{\mathbb{R}^\times}{\mathbb{R}^\times_{>0}} \simeq \mathbb{Z}/2\mathbb{Z}.$$

2. When $k = \mathbb{F}_q$ is a finite field the Brauer group vanishes: $\mathrm{Br}(\mathbb{F}_q) = 0$. By the lemma 2) it is enough to show that $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$ for every $r > 0$. Recall the following facts:
   - The group $\mathrm{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)$ is cyclic, generated by the frobenius $a \mapsto a^q$. In particular the norm has value
   $$N(a) = a^{\frac{q^r-1}{q-1}};$$
   - The multiplicative group $\mathbb{F}_{q^r}^\times$ is cyclic.

   Let ω be a generator of $\mathbb{F}_{q^r}^\times$, its norm $N(\omega)$ by the above calculation has order $q-1$ and therefore generates $\mathbb{F}_q$. This shows N is surjective and by lemma 3) we deduce $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$.

1.1. The path to $K|\mathbb{Q}_p$

The main goal of this section is proving that if $K|\mathbb{Q}_p$ is a p-adic number field then its Brauer group is $\mathrm{Br}(K) = \mathbb{Q}/\mathbb{Z}$.

Definition 1.1.2 (Discretely valuated fields)

A *discretely valuated field* is a field equipped with a discrete valuation $v : K \to \mathbb{Z} \cup \{\infty\}$, i.e. it holds
- $v(x) = \infty \iff x = 0$;
- $v(xy) = v(x)v(y)$;
- $v(x+y) \geq \min\{v(x), v(y)\}$.

Moreover we say K is *complete* if it complete with respect to the norm $\|x\|_v = e^{-v(x)}$. The *valuation ring* of K is defined as

$$\mathcal{O}_K := \{x \in K : v(x) \geq 0\},$$

it is a discrete valuation ring with maximal ideal $M_K := \{x \in K : v(x) > 0\}$. The *residue field* of K is the quotient $k(v) := \mathcal{O}_K/M_K$.

Fact

The ideal M_k is principal and it is generated by any element $\pi \in K$ with valuation $v(\pi) = 1$, such an element is called a *uniformiser* for $\mathcal{O}_K$. Since M_K is maximal, $k(v) = \mathcal{O}_K/(\pi)$ is indeed a field.

Fact

The field K is complete if and only if its valuation ring $\mathcal{O}_K$ is a complete DVR, that is

$$\mathcal{O}_K \simeq \varprojlim_i \frac{\mathcal{O}_K}{M_K^i}.$$

Example 1.1.2

1. The p-adic field $\mathbb{Q}_p$ is complete as $\mathbb{Q}_p = \mathrm{Frac}(\varprojlim \mathbb{Z}/p^i\mathbb{Z})$.
2. The Laurent series $k((t))$ form a complete field as $k((t)) = \mathrm{Frac}(\varprojlim k[\![t]\!]/(t^i))$.

Fact

If (K, v) is a complete discretely valuated field and $L|K$ is a finite separable extension, the valuation can be extended uniquely to L via

$$w = \frac{1}{[L:K]} v \circ N_{L|K} : L^\times \to \frac{1}{[L:K]}\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}.$$

The discretely valuated field (L, w) is complete itself and its valuation ring $\mathcal{O}_L$ is the integral closure of $\mathcal{O}_K$ inside L.

Definition 1.1.3 (ramification and residue class)

If $L|K$ is a finite separable extension of complete discretely valuated fields, the associated *ramification index* is $e(L|K) := [w(L^\times) : v(K^\times)]$ and the *residue class degree* is $f(L|K) := [k(w) : k(v)]$.

We say $L|K$ is *unramified* if $k(w)|k(v)$ is separable and $e = 1$; and $L|K$ is *totally ramified* if $f = 1$.

From now on we assume that K is a complete discretely valuated field.

Facts

- When $k(w)|k(v)$ is separable it holds $[L:K] = e(L|K) \cdot f(L|K)$.
- If $L|K$ is finite and unramified there exists a unique extension $L \supset M \supset K$ such that $M|K$ is unramified and $L|M$ is totally ramified.
- *Towers property:* If both $L|K, L'|L$ are finite and unramified, $L'|K$ is also finite and unramified.
- *Shift property:* If $L|K$ is finite and unramified, $K'|K$ is finite and separable; the extension $LK'|K'$ is finite and unramified.

Definition 1.1.4 (Unramified closure)

We set the maximal unramified extension of K as

$$K_{\mathrm{nr}} = \bigcup \{\overline{K} \supset L \supset K : L|K \text{ is finite and unramified}\}.$$

The maximal unramified extension K_{nr} is a discrete valuated field, with valuation extending that of K. Its residue field is the separable closure $k(v)_s$, however in general it is not a complete field.

Suppose $L|K$ is a finite Galois extension (this can be generalized to infinite Galois extensions) and let $G = \mathrm{Gal}(L|K)$. The action of G on L is compatible with the valuation (this is a consequence of the uniqueness of valuation extensions), meaning $\mathcal{O}_L$ and M_L are G-invariant. Every element $\sigma \in G$ induces an automorphism

$$\overline{\sigma} : \frac{\mathcal{O}_L}{M_L} \to \frac{\mathcal{O}_K}{M_K}, \qquad \overline{x} \mapsto \overline{\sigma(x)},$$

giving a homomorphism on the Galois group of the residue fields extension

$$\rho : \mathrm{Gal}(L|K) \to \mathrm{Gal}(k(w)|k(v)).$$

By Hensel's lemma this is surjective, moreover its kernel is given by the *inertia subgroup* $I(L|K)$ and it is known that $|I(L|K)| = e(L|K)$. In particular when the extension is unramified (i.e. $e = 1$) the map ρ is an isomorphism.

Notice that by a similar argument if $L|K$ is a finite unramified extension then $L|K$ is Galois if and only if the residue field extension $k(w)|k(v)$ is Galois.

Example 1.1.3

Suppose the residue field $k(v)$ is finite. Since finite extension on finite fields are Galois, every finite unramified extension $L|K$ is a cyclic Galois extension. This means that for every $n \mid [L : K]$ there is a unique unramified subextension $L \supset K_n \supset K$ of degree $[K_n : K] = n$. In other words if we fix a closure $\overline{K}$ for every n there is a unique unramified Galois extension $\overline{K} \supset K_n \supset K$ of degree n.

Definition 1.1.5 (Units and principal units groups)

The *units group* of K is the subgroup $\mathcal{U}_K := \{a \in K^\times : v(a) = 0\}$. The *$i$-th group of principal units* is $\mathcal{U}_K^{(i)} := \{a \in K^\times : v(a-1) \geq i\}$.

There is an exact sequence of groups

$$1 \longrightarrow \mathcal{U}_K \longrightarrow K^\times \xrightarrow{v} \mathbb{Z} \longrightarrow 1,$$

the units group can be interpreted as the kernel of the valuation map. Moreover there is a surjection $\mathcal{U}_K \twoheadrightarrow k(v)^\times$ given by $u \mapsto u \bmod M_K$ and the principal units group can be viewed as its kernel:

$$\mathcal{U}_K^{(1)} = \ker(\mathcal{U}_K \longrightarrow k(v)^\times).$$

Notice that $\bigcap_i \mathcal{U}_K^i = \{1 + a \in K^\times : v(a) \geq i \text{ for each } i \geq 1\} = \{1\}$. The map $\mathcal{U}_K^{(i)} \to M_K^i$ given by $u \mapsto u \bmod M_K^{i+1}$ induces an isomorphism

$$\frac{\mathcal{U}_K^{(i)}}{\mathcal{U}_K^{(i+1)}} \xrightarrow{\sim} \frac{M_K^i}{M_K^{i+1}} \simeq \frac{\pi^i \mathcal{O}_K}{\pi^{i+1}\mathcal{O}_K} \simeq \frac{\mathcal{O}_K}{(\pi)} \simeq k(v).$$

We have therefore constructed a filtration (consisting of an infinite number of elements)

$$\mathcal{U}_K \underset{k(v)^\times}{\supset} \mathcal{U}_K^{(1)} \underset{k(v)^+}{\supset} \mathcal{U}_K^{(2)} \underset{k(v)^+}{\supset} \ldots \supset \{1\},$$

this filtration can be useful to prove properties of the multiplicative group $K^\times$.

Example 1.1.4

Consider a finite p-adic extension $K|\mathbb{Q}_p$. The residue field $k(v)$ is a finite extension of $\mathbb{F}_p$, hence it is of the form $k(v) = \mathbb{F}_q$ where $q = p^r, r = [K : \mathbb{Q}_p]$. The exact sequence

$$1 \longrightarrow \mathcal{U}_K^{(1)} \longrightarrow \mathcal{U}_K \longrightarrow k(v)^\times \to 1$$

splits by lifting $k(v)^\times$ to the $q-1$ roots of unity in $\mathcal{U}_K$ via $k(v)^\times \to \mu_{q-1}$. This gives the decomposition

$$K^\times \simeq \mathbb{Z} \times \mathcal{U}_K \simeq \mathbb{Z} \times \mu_{q-1} \times \mathcal{U}_K^{(1)}.$$

It can be shown that $\mathcal{U}_K^{(1)}$ itself can be decomposed as $\mathcal{U}_K^{(1)} \simeq \mu_{p^s} \times \mathbb{Z}_p^r$, where p^s is the number of p-power roots of unity in K.

We deduce $K^\times / K^{\times n}$ is a finite abelian group.

1.2. Computing $\mathrm{Br}(K)$ in the finite residue field case

The goal of this section is to show that whenever K is a complete discretely valuated field with finite residue field (as in a finite extension $K|\mathbb{Q}_p$), the Brauer group is given by $\mathrm{Br}(K) \simeq \mathbb{Q}/\mathbb{Z}$. We divide our goal in two steps: proving $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$ and proving $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathrm{Br}(K)$.

Lemma 1.2.2

Let $L|K$ be a finite unramified Galois extension and let $G = \mathrm{Gal}(L|K)$. There is a split short exact sequence

$$0 \longrightarrow H^2(G, \mathcal{U}_L) \longrightarrow H^2(G, L^\times) \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

Proof: The short exact sequence

$$1 \longrightarrow \mathcal{U}_L \longrightarrow L^\times \xrightarrow{v} \mathbb{Z} \longrightarrow 0$$

splits via $s : \mathbb{Z} \to L^\times, 1 \mapsto \pi$ where π is a uniformiser of $\mathcal{O}_L$. Since the extension is unramified, we can choose $\pi \in K$, so that the section s is G-equivariant. Thus s induces a map on cohomology (that we still denote s) and the exact sequence

$$0 = H^1(G, \mathbb{Z}) \longrightarrow H^2(G, \mathcal{U}_L) \longrightarrow H^2(G, L^\times) \longrightarrow H^2(G, \mathbb{Z}) \longrightarrow 0$$

also splits via the section $s : H^2(G, \mathbb{Z}) \to H^2(G, L^\times)$ (indeed, the existence of the section guarantees surjectivity of the right-hand map). It remains to justify that $H^1(G, \mathbb{Z}) = 0$: since the G-action on $\mathbb{Z}$ is trivial we have $H^1(G, \mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Z})$ and every homomorphism from a finite group to $\mathbb{Z}$ is trivial.

It remains to identify $H^2(G, \mathbb{Z})$ with $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$. Consider the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0,$$

the associated long sequence in cohomology gives

$$0 = H^1(G, \mathbb{Q}) \longrightarrow H^1(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} H^2(G, \mathbb{Z}) \longrightarrow H^2(G, \mathbb{Q}) = 0.$$

As we've seen since $\mathbb{Q}$ is divisible its finite group cohomology vanishes in positive dimension. Since the G-action on $\mathbb{Q}/\mathbb{Z}$ is trivial we have $H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ and we conclude. □

Lemma 1.2.3

Let G be a finite group and let (A_j, φ_j) be an inverse system of G-modules indexed by $\mathbb{Z}_{>0}$ (in particular the transition maps are the $\varphi_j : A_{j+1} \to A_j$). For any i, if $H^i(G, A_1) = 0$ and $H^i(G, \ker \varphi_j) = 0$ for every j, then $H^i(G, \varprojlim_{\leftarrow} A_j) = 0$.

Lemma 1.2.4 (Additive Hilbert's 90)

If K is a field and $L|K$ is a finite Galois extension with group G, then $H^i(G, L^+) = 0$ for every $i > 0$.

In particular from Lemma 1.2.3 we deduce $H^i(k, k_s) = 0$ for every $i > 0$.

Proof: Recall the *normal basis theorem*: there is an element $a \in L$ such that the set $\{\sigma a : a \in G\}$ forms a k-basis for L. We deduce there is an isomorphism of G-modules

$$k^+ \otimes_{\mathbb{Z}} \mathbb{Z}[G] \to L^+, \qquad \lambda_i \otimes \sigma_i \mapsto \lambda_i \sigma_{i(a)}.$$

Moreover there is a non-canonical isomorphism $k^+ \otimes_{\mathbb{Z}} \mathbb{Z}[G] \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], k^+)$ of k-vector spaces since they have the same (finite) dimension. The latter is the coinduced-G-module $M^G(k^+)$ and therefore $H^i\big(G, M^G(k^+)\big) = 0$. □

Proposition 1.2.1

Let $L|K$ be a finite unramified Galois extension with Galois group G. Let κ and λ denote the residue fields of K and L, respectively. The reduction map $\mathcal{U}_L \to \lambda^\times$ induces an isomorphism $H^2(G, \mathcal{U}_L) \xrightarrow{\sim} H^2(G, \lambda^\times)$.

Proof:

□

Corollary 1.2.1

We have a split short exact sequence

$$0 \longrightarrow \mathrm{Br}(\lambda|\kappa) \longrightarrow \mathrm{Br}(L|K) \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

In particular when κ is finite if we let $n = [L : K]$ then $\mathrm{Br}(L|K) \simeq \mathbb{Z}/n\mathbb{Z}$.

Proof: By Lemma 1.2.2 and Proposition 1.2.1 we get the desired sequence. When κ is finite, since the extension $\lambda|\kappa$ is itself finite (recall that $\mathrm{Gal}(\lambda|\kappa) \simeq \mathrm{Gal}(L|K)$ in our case) we have $\mathrm{Br}(\lambda|\kappa) = 0$. Moreover $n = [L : K] = |G|$ and therefore $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$. □